

On the Kinematic Error in Harmonic Drive Gears

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Harmonic drive gears are widely used in space applications, robotics, and precision positioning systems because of their attractive attributes including near-zero backlash, high speed reduction ratio, compact size, and small weight. On the other hand, they possess an inherent periodic positioning error known as kinematic error responsible for transmission performance degradation. No definite understanding of the mechanism of kinematic error as well as its characterization is available in the literature. In this paper, we report analytical and experimental results on kinematic error using a dedicated research Harmonic Drive Test Apparatus. We first show that the error referred to in the literature as kinematic error actually consists of a basic component, representing "pure" kinematic error, colored with a second component resulting from inherent torsional flexibility in the harmonic drive gear. The latter component explains the source of variability in published kinematic error profiles. The decomposition of the kinematic error into a basic component and a flexibility related component is demonstrated experimentally as well as analytically by matching a mathematical model to experimental data. We also characterize the dependence of the kinematic error on inertial load, gear assembly, and rotational speed. The results of this paper offer a new perspective in the understanding of the mechanism of kinematic error and will be valuable in the mechanical design of harmonic drive gears as well as in the dynamic modeling and precision control of harmonic drive systems. [DOI: 10.1115/1.1334379]

Keywords: Harmonic Drive Gears, Kinematic Error, Flexibility, Load/Assembly/Speed Dependence

1 Introduction

The harmonic drive gear is a mechanism that is ideal for use in applications where precise positioning with high gear reduction is required. The device is compact in that large torques may be produced by relatively small motors because of its velocity reduction and torque amplification properties. This capability permits the use of small motors and harmonic drives in applications where much larger motors and bulky gear boxes would be otherwise required. This makes them ideal for robotic applications since they can be directly mounted at the joint along with the motor. The drive is designed such that several teeth are engaged at any given time making backlash virtually zero. Hence they are popular in applications requiring precision positioning such as robots that manufacture precision components (e.g. printed circuit boards), and precise measuring devices. They are widely used in the semiconductor industry for laser mirror positioning. Possible military applications include use in missile fin actuation systems and in fine positioning mechanisms for laser redirection. Harmonic drives are making headway in commercial, industrial, and military applications.

A harmonic drive is composed of the components identified in Fig. 1(a). The wave-generator is an elliptically shaped steel core surrounded by a flexible race bearing. The circular spline is a rigid steel ring with teeth machined into the inner circumference. The flexible spline (or flexspline) is a thin-walled flexible cup having two fewer teeth on its outer rim than on the inner rim of the circular spline. Upon assembly, the wave-generator is inserted into the flexspline cup which assumes an elliptical shape at that

end. The other end, however, is circular in shape and is attached to the output shaft. The circular spline teeth then mesh with the flexspline teeth at the major axis of the ellipse defined by the wave-generator. A fully assembled harmonic drive is shown in Fig. 1(b). The most common configuration for the harmonic drive is the speed reduction/torque magnification arrangement. This mode of operation usually consists of the wave-generator as the input port, the flexspline as the output port, and the circular spline fixed to ground and held immobile. In this configuration, the wave-generator rotation corresponds to the motor angle input while the rotation of the flexspline in the opposite direction corresponds to the load angle output.

The theory underlying the operation of harmonic drive gears was developed during the mid-1950s [2,3]. The technology has advanced since that point, but research into the theoretical aspects of the transmissions and their inherent inaccuracies has not been extensive. Most of the research on this topic has been performed by engineers in the former Soviet Union [4–6], Canada [7–9], Japan [10–12], and the United States [13–20].

Research in the past has been primarily in the nonlinear transmission attributes of harmonic drives including the kinematic error, flexibility, and friction, and design attributes including tooth stresses and deformation. The kinematic error is defined as the deviation between the expected output position and the actual output position and may be represented by the following equation:

$$\tilde{\theta} = \frac{\theta_m}{N} - \theta_l, \quad (1)$$

where θ_m is the rotation of the motor shaft attached to the wave-generator, N is the gear reduction ratio, and θ_l is the rotation of the output shaft connected to the flexspline or circular spline as the case may be. Figure 2 shows a typical kinematic error waveform recorded using the dedicated harmonic drive setup described in Appendix A at a motor speed of 300 rpm and at no load. As we

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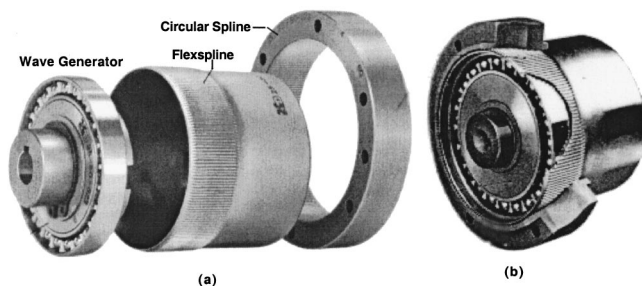


Fig. 1 (a) Exploded and (b) assembled view of harmonic drive gear transmission (Harmonic Drive Technologies [1])

can observe, there is a basic harmonic occurring at twice the wave generator rotation and there are higher frequency components of the error superimposed on the basic form.

Kinematic error, though small in magnitude, is periodic in nature and acts as an exciter and hence has undesirable vibration effects. These vibrations become dominant at higher speeds, especially at the resonant frequencies. They serve as an energy sink and have been known to produce dramatic torque losses and velocity fluctuations [13,7] affecting precision positioning and tracking operations. Thus the kinematic error, in combination with the nonlinear friction and flexibility effects, plays an important role in transmission performance.

The literature, however, still lacks a precise characterization of the mechanism responsible for the kinematic error. First, the origin of kinematic error is not known precisely. Emel'yanov et al. [14] carried out a mathematical analysis regarding the source of kinematic error and concluded that the error is due to assembly and physical imperfections on the three principal elements of a harmonic gear drive. They also pointed out that their results revealed the presence of harmonics which were periodic in nature occurring at twice the wave generator rotation. Hsia [16] and Ramson [17] both proposed that the kinematic error in harmonic drives is due to the deformation of the flexspline when it takes on the shape of the wave generator. In fact, Hsia [18] showed that the kinematic error is a result of inherent operating principles of the drive irrespective of the assembly errors. However, the effect of deformation of flexspline cup in the longitudinal direction is not taken into consideration in his derivation. Nye and Kraml [15] experimentally studied kinematic error and proposed the existence

of “well-behaved” and “poorly-behaved” kinematic error forms which distinguish between the fundamental error form on one hand and the fundamental form superimposed with higher frequency components on the other. An extensive experimental analysis of the kinematic error present in harmonic drives was carried out by Tuttle and Seering [15]. Their research led to the belief that the primary contributor to the kinematic error component was the meshing of the gear teeth and that the periodic nature of the signal would be explained as being a function of the tooth meshing frequency. This conclusion actually agrees with Emel'yanov et al. [4] in that the harmonic with the strongest response corresponds to a frequency of twice the wave generator rotation.

The available literature indicates that seemingly identical harmonic gear drive units will produce different error signatures, a phenomenon attributed to particular physical gear imperfections due to manufacturing and assembly. This suggests that kinematic error magnitudes, harmonics, and periodicity need to be determined experimentally. In addition, all of the researchers studying transmission attributes of harmonic gear drives have observed that positional error, as well as other properties, is affected by environmental conditions. On the other hand, none of the studies in the literature considered the influence of speed, load, and stiffness of the drive on kinematic error.

Using a dedicated research harmonic drive test apparatus, we present in this paper experimental and analytical results that shed new light toward a better understanding of the mechanism of kinematic error as well as its characterization. We first propose that the kinematic error known in the harmonic drive manufacturers' literature as well as in the scientific literature is actually comprised of a basic, “pure,” component and a second component due to inherent torsional flexibility. We therefore design a very slow speed experiment to minimize the effect of flexibility, hence, giving the basic form of the kinematic error. We then devise a mathematical model that takes into account the basic kinematic error as well as torsional flexibility and show that predicted simulation and experimental data of the kinematic error match quite well under different experimental conditions. This proves that our decomposition proposal is quite reasonable and gives an explanation for the source of variability and apparent “randomness” of the reported kinematic error profiles in the literature. Indeed, it is the flexibility-related component of the kinematic error, among other predictable factors as we will show later, that is responsible for this variability. We also report on extensive experimental analysis studying the dependence of the kinematic error on inertial load, gear assembly, and angular velocity. When appropriate, we compare experimental data to simulation data.

This paper is organized as follows: Section 2 discusses kinematic error in harmonic drives and develops its decomposition into a pure component and a torsional flexibility-induced component. Section 3 illustrates the factors influencing the kinematic error profiles, namely, external load, depth of penetration of the wave generator in the flexspline, and the angular velocity. Finally, Section 4 presents conclusions of this study.

2 Decomposition of the Kinematic Error

The profile of the kinematic error in harmonic drives has been reported to vary from drive to drive, and even seemingly similar harmonic drives can produce different kinematic error signatures. Even though many of the factors influencing the profile of the kinematic error have not been fully explored in the literature, it is generally accepted that the error has a fundamental component corresponding to twice the frequency of the wave generator, colored with higher frequency components. In this section we present a more precise characterization and propose that the kinematic error is mostly dominated by two major components. The first component, $\tilde{\theta}_p$, is a basic component that is “pure” kinematic error resulting from the kinematic structure of the harmonic drive. The second component, $\tilde{\theta}_s$, is mostly due to the stiffness proper-

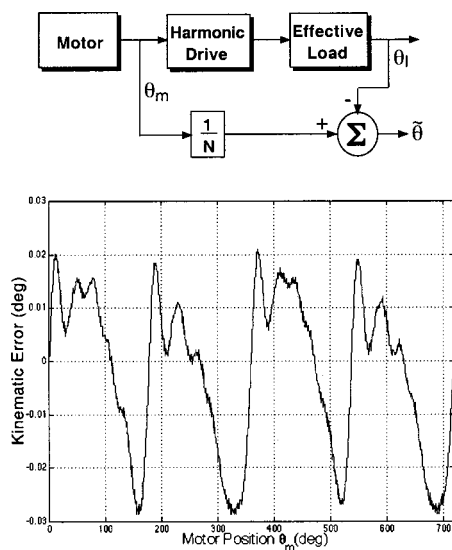


Fig. 2 Typical profile of the kinematic error

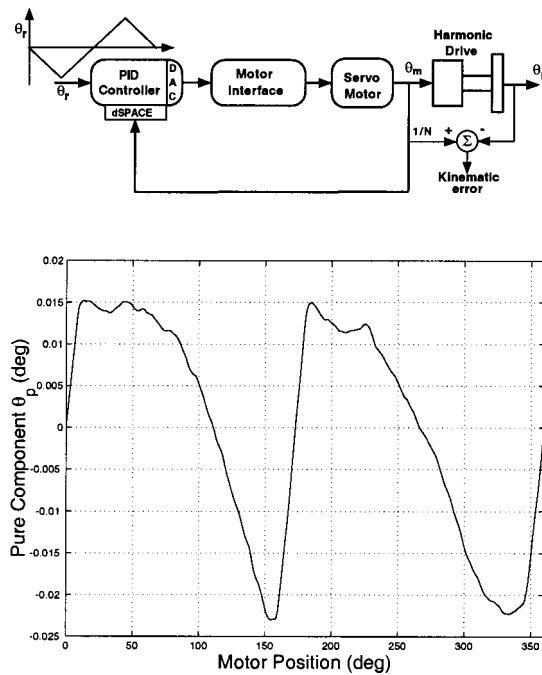


Fig. 3 Experimental measurement of basic component of the kinematic error

ties of the drive. A typical steady-state stiffness curve is given in Appendix B in Fig. 13 which shows that the stiffness of the harmonic drive is particularly low at low torques.² This in turn implies that when the harmonic drive is in motion, relative motion between the wave generator (input) and the flexpline (output) will be generated, coloring the basic component of the kinematic error. Consequently, the expression of the kinematic error introduced in (1) could actually be decomposed into

$$\tilde{\theta} = \tilde{\theta}_p + \tilde{\theta}_s. \quad (2)$$

We therefore design special experiments to measure the basic component of the kinematic error $\tilde{\theta}_p$, and illustrate the stiffness related component of the error $\tilde{\theta}_s$ by comparing experimental data with simulation data.

2.1 Basic Component of Kinematic Error. To get the kinematic error in its pure form, experiments should be carried out at very low speed to avoid the excitation of the vibrations due to flexibility. This is very important, since the stiffness of the drive as measured experimentally is low for small deflections, as pointed out earlier. Rotating the wave generator at a speed of around 10 rpm is found to be low enough for suppressing the effects of flexibility for our drive. This speed may vary from drive to drive depending on the drive stiffness and was determined in our case by running experiments at successively decreasing speeds and selecting the one at which the kinematic error waveform shows no significant variation. However, running the wave generator at such low speeds is not a simple task. Generally, one has to apply low torque input in order to achieve low speeds. However, with low torque input, the effects of friction in the drive, especially stiction, become dominant, resulting in intermittent motion or sticking of the drive. To overcome this problem, a special experiment was designed wherein a PID controller is developed to make the motor position follow a triangular input reference waveform. Figure 3 shows the block diagram of the setup. The frequency and the amplitude of the triangular reference motor

²This is sometimes referred to as "soft-windup" phenomenon (Harmonic Drive Technologies [1])

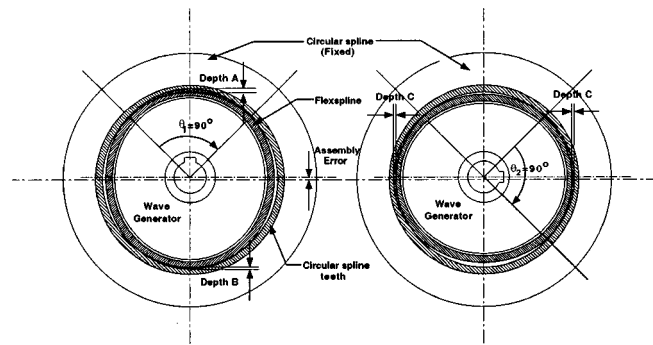


Fig. 4 Mechanism of occurrence of kinematic error due to assembly errors

position is adjusted to 0.01 Hz and 1440 deg to get the desired slow speed of 9.6 rpm with smooth motion. Proper tuning of controller gains avoided sticking, thereby producing the smooth motion of the drive at such low speed. By controlling the motor speed in the way mentioned above, the kinematic error component $\tilde{\theta}_p$ is obtained and is shown in Fig. 3 for one wave generator revolution. As seen in the figure, the profile of the kinematic error for the drive under consideration consists predominantly of a fundamental harmonic occurring at a frequency of two cycles per wave generator rotation.

Extensive experimentation carried out as part of this work indicates that the source of the fundamental frequency of the above kinematic error profile can be traced to assembly inaccuracies as indicated by Emel'yanov et al. [4]. The small amplitude, high frequency, components superimposed on this fundamental are believed to occur due to the teeth placement errors. We propose the following explanation of the mechanism of occurrence of the error. The slight misalignment of the circular spline due to assembly errors and/or shaft deflection would cause the flexpline teeth to move deeper (depth A) into the circular spline on one side of the major axis of the wave generator ellipse than on the opposite side (depth B), as shown in Fig. 4. Higher depth of meshing on one side results in moving the load (for the same motor revolution) faster when the wave generator traverses that side (angle $\theta_1 = 90^\circ$ in Fig. 4) which in turn results into the negative slope on the kinematic error waveform (load leads the motor). During the next 90 degree (angle θ_2) of wave generator rotation the load now lags the motor producing positive slope on the kinematic error waveform. This serves as a basic mechanism of occurrence of the fundamental harmonic of the kinematic error based on assembly errors. The basic mechanism remains the same for other assembly errors.

2.2 Flexibility Induced Component of the Kinematic Error. In this section, we show that the kinematic error measured under typical motion of the harmonic drive is composed of the pure component discussed in the previous section as well as a second component induced by the stiffness of the harmonic drive gear. In order to justify this argument, we develop a mathematical dynamic model of the harmonic drive system that takes into consideration the pure kinematic error as well as the flexibility of the drive. The experimentally measured kinematic error was compared to the profile generated by simulation and good agreement was observed, indicating that our hypothesis of the decomposition of the kinematic error is very reasonable.

In order to develop a mathematical dynamic model of the Harmonic Drive Test Apparatus of Fig. 12 in Appendix A, its schematic (shown horizontally for convenience) is given in Fig. 5. The figure illustrates the parameters of the system, whose values are displayed in Table 1. The parameters were computed from the geometry of the system and by performing experiments at differ-

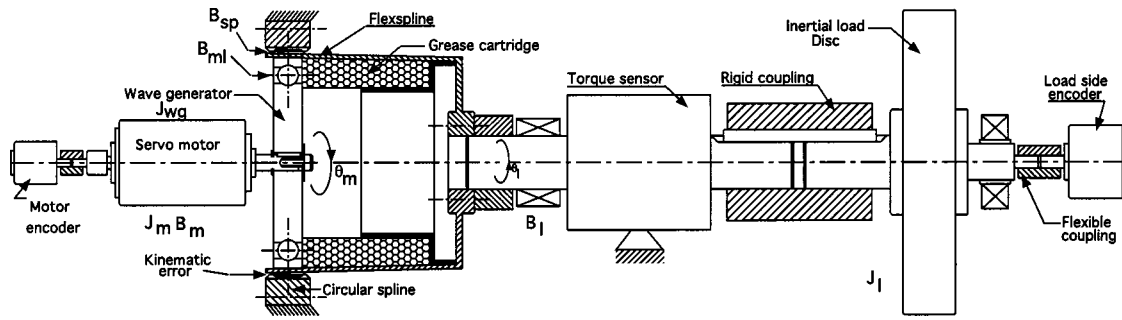


Fig. 5 A schematic diagram of the Harmonic Drive Test Apparatus

ent speeds to obtain estimates on viscous damping factors and constant frictional torque. To account for the effects of flexibility, we assume that a torsional spring exists between input and output side of the drive characterized by a constant stiffness coefficient. This is justified by the fact that, even though stiffness is nonlinear and is hysteretic (because of combined effect of flexibility and friction), the motion of the drive in all the experiments conducted in this study on kinematic error is unidirectional and results in small torsional displacement corresponding to a small interval with the constant slope of 7160 Nm/rad along the stiffness curve as shown in Fig. 13.

The kinetic energy of the system consists of the energy components due to moving load-side and motor-side elements,

$$T = \frac{1}{2} J_m \dot{\theta}_m^2 + \frac{1}{2} J_l \dot{\theta}_l^2, \quad (3)$$

where J_m and J_l are the lumped moment of inertia from the motor and load side, respectively, and θ_m and θ_l are the independent generalized coordinates of the motor and load respectively. The potential energy due to stiffness is given by

$$V = \int_{\theta_l - \theta_m/N + \tilde{\theta}_p}^0 K \sigma d\sigma, \quad (4)$$

where $K\sigma$ is a torsional torque for a rotational deflection σ . Note that the lower limit in (4) takes into consideration the basic component of the kinematic error $\tilde{\theta}_p$ which, as we have shown, is dependent on the motor position θ_m . Viscous damping in the system can be classified into three categories. First, there is damping on the motor side, B_m , and load side, B_l , which are due to all the bearings on the motor and load sides and the damping due to misalignment of flexible coupling between the different components. Second, there is damping B_{ml} which accounts for the resistance to the motion of the wave generator in the flexspline cup. Finally, there is damping B_{sp} which acts in conjunction with the stiffness of the drive. It follows that the Rayleigh dissipation function can be expressed as:

$$D = \frac{1}{2} B_m \dot{\theta}_m^2 + \frac{1}{2} B_l \dot{\theta}_l^2 + \frac{1}{2} B_{ml} (\dot{\theta}_m + \dot{\theta}_l)^2 + \frac{1}{2} B_{sp} \left[\dot{\theta}_l - \frac{\dot{\theta}_m}{N} + \dot{\tilde{\theta}}_p \right]^2. \quad (5)$$

Consequently, the Lagrange equations of motion are given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_m} \right) - \frac{\partial T}{\partial \theta_m} + \frac{\partial V}{\partial \theta_m} + \frac{\partial D}{\partial \dot{\theta}_m} = \tau_m$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_l} \right) - \frac{\partial T}{\partial \theta_l} + \frac{\partial V}{\partial \theta_l} + \frac{\partial D}{\partial \dot{\theta}_l} = 0.$$

Substituting for various expressions in these equations and simplifying we obtain

$$J_m \ddot{\theta}_m + K \left(\theta_l - \frac{\theta_m}{N} + \tilde{\theta}_p \right) \left[-\frac{1}{N} + \frac{d\tilde{\theta}_p}{d\theta_m} \right] + B_m \dot{\theta}_m + B_{ml} (\dot{\theta}_m + \dot{\theta}_l) + B_{sp} \left(\dot{\theta}_l - \frac{\dot{\theta}_m}{N} + \dot{\tilde{\theta}}_p \right) \left[-\frac{1}{N} + \frac{d\tilde{\theta}_p}{d\theta_m} \right] = \tau_m, \quad (6)$$

$$J_l \ddot{\theta}_l + K \left(\theta_l - \frac{\theta_m}{N} + \tilde{\theta}_p \right) + B_l \dot{\theta}_l + B_{ml} (\dot{\theta}_m + \dot{\theta}_l) + B_{sp} \left(\dot{\theta}_l - \frac{\dot{\theta}_m}{N} + \dot{\tilde{\theta}}_p \right) = 0. \quad (7)$$

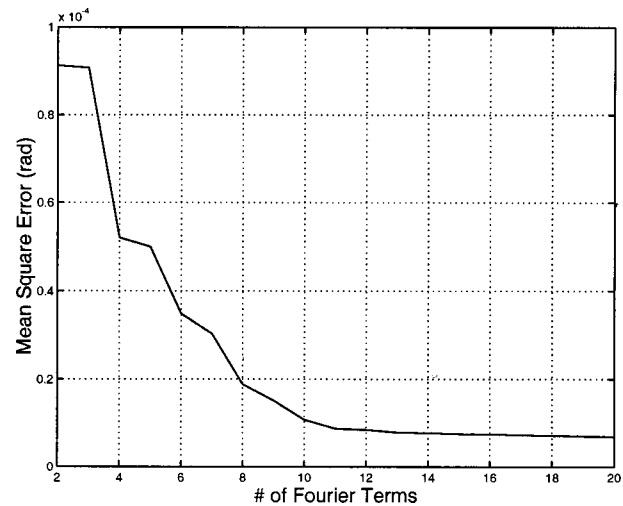
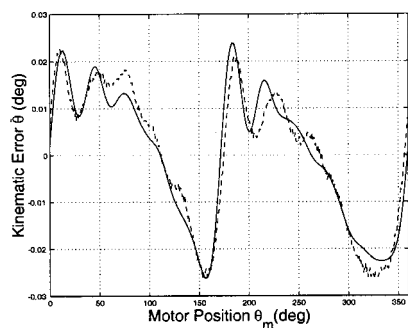


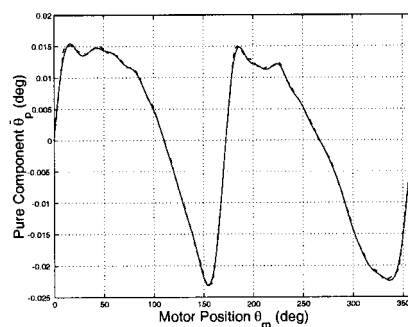
Fig. 6 Mean Square Error (MSE) between simulated and experimental profiles

Table 1 Dynamic system parameters used for simulation

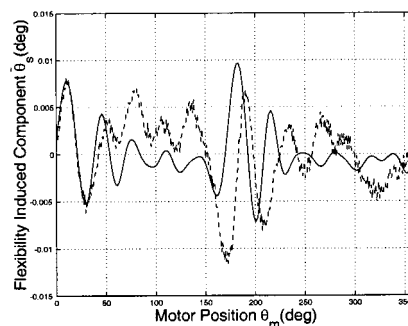
Parameter	Numerical Value
Motor Inertia (J_{motor})	$2.9 \times 10^{-4} \text{ kgm}^2$
Wave-generator Inertia (J_l)	$1.6 \times 10^{-4} \text{ kgm}^2$
Inertial Load (J_{load})	$5.0 \times 10^{-2} \text{ kgm}^2$
Motor Bearing Damping (B_{mot})	$1.7 \times 10^{-4} \text{ Nm/s}$
Wave-generator Bearing damping (B_l)	$1.3 \times 10^{-5} \text{ Nm/s}$
Load Bearing Damping (B_{load})	$5.0 \times 10^{-4} \text{ Nm-s}$
Gear Ratio (N)	50
Stiffness of Spring (load side) K	7160 Nm/rad.



(a) Typical kinematic error profile



(b) Pure component



(c) Flexibility-induced component

Fig. 7 Decomposition of the kinematic error: Experimental (dashed) and simulated (solid)

Note that in the dynamic equation of the drive system (6)–(7), the pure component of the kinematic error $\tilde{\theta}_p$ and its derivative $\dot{\tilde{\theta}}_p$ have to be known explicitly at any given time instant. To derive an expression of $\tilde{\theta}_p$ as a function of the motor angle θ_m , a Fourier series expansion is used to analytically express the pure kinematic error experimental data as a function of θ_m as follows:

$$\tilde{\theta}_p = \frac{a_0}{2} + \sum_{n=1}^k [a_n \cos(n\theta_m) + b_n \sin(n\theta_m)], \quad (8)$$

where,

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \tilde{\theta}_p(\theta_m) \cos(n\theta_m) d\theta_m,$$

Table 2 Fourier coefficients used in the kinematic error simulation

n	$a_n(\text{deg})$	$b_n(\text{deg})$
0	−0.00897	0.0000
1	0.00243	−0.00158
2	−0.00086	0.01692
3	0.00325	0.00177
4	0.00025	−0.00060
5	0.00082	0.00003
6	−0.00131	0.00382
7	0.00117	0.00241
8	0.00120	0.00079
9	0.00126	0.00256
10	0.00120	0.00103
11	−0.00359	0.00124
12	−0.00090	−0.00060
13	0.00017	−0.00049
14	−0.00065	−0.00039
15	0.00042	−0.00010

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \tilde{\theta}_p(\theta_m) \sin(n\theta_m) d\theta_m.$$

Constants a_n and b_n are determined from experimental data by numerical integration. Using this approximation to the experimental profile, Fig. 6 shows the Mean Square Error (MSE, computed using Parseval's relation) versus the number of terms of Fourier series (k in Eq. (8)). We observe that fifteen terms are sufficient to closely model the experimental data. Figure 7(b) simultaneously displays experimental data of the pure kinematic error (originally presented in Fig. 3) along with a fifteen-term Fourier series expansion whose coefficients are displayed in Table 2.

The main result of this section is now illustrated in Fig. 7(a), which displays a typical experimental kinematic error profile (corresponding to a motor velocity of 310 rpm) and simulation data using the equations of motion (6)–(7). In Fig. 7(c), the stiffness-induced component of the kinematic error $\tilde{\theta}_s$ is displayed from experimental and simulation data. The closeness of experimental and simulation results in Fig. 7 proves that the decomposition proposal of kinematic error into these two components is reasonable. It also implies that the flexibility-induced component in the kinematic error is mostly the source of the high-frequency components of kinematic error reported in the literature.

3 Factors Influencing the Kinematic Error

The previous decomposition of the kinematic error into a pure component $\tilde{\theta}_p$ and a flexibility-induced component $\tilde{\theta}_s$ will be further analyzed in this section by outlining our findings on some of the important factors that influence one or both of these components. The observations reported in these sections are from tests we conducted on the Harmonic Drive Test Apparatus of Fig. 12 in Appendix A. When appropriate, we made comparison with simulation results using our dynamic model derived in the previous section.

3.1 Load-Dependence. We present results of experiments carried out to test the load-dependence of the pure component of the kinematic error $\tilde{\theta}_p$. The procedure for the experiments is similar to that designed for obtaining the basic form, except that now the output shaft of the harmonic drive is loaded with a torsional load applied by means of a weight and pulley arrangement, as shown in Fig. 8 (note that the inertia disc in this figure corresponds to the inertial load pointed out in Fig. 12 and Fig. 5). It was observed that as the load increases, the kinematic error (peak-to-peak) amplitude decreases. Similarly, the profile of the kinematic error also shows a corresponding change but is not reported in this study. Figure 9 shows a plot (fitted experimental data points) of kinematic error peak-to-peak amplitude as the load is varied. As the figure shows, the amplitude is very sensitive to load

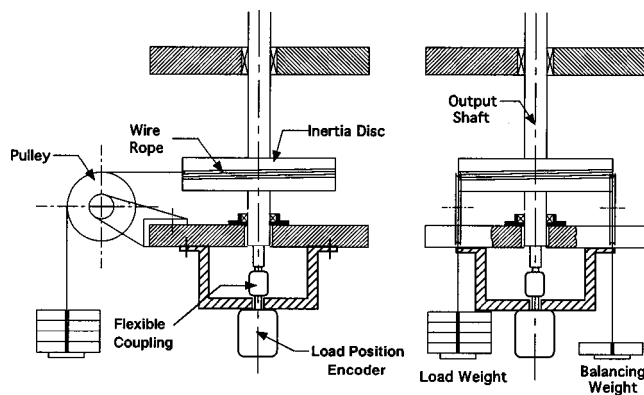


Fig. 8 Loading arrangement

variations at smaller loads. This can be attributed to the soft-windup phenomenon (Harmonic Drive Technologies [1]). At higher loads, however, the peak-to-peak amplitude tends to become constant. Thus the sensitivity of the kinematic error (peak-to-peak magnitude as well as profile shape) to inertial load further complicates the problem of characterization and modeling of the kinematic error. This is because in a typical motion trajectory, the dynamic load (induced by a variation in motion acceleration) is variable. Consequently, for a given motion trajectory, the pure component of the kinematic error is not “stationary;” rather, it belongs to a family of profiles, each of which corresponds to a particular value of the effective load applied at the harmonic drive.

3.2 Gear Assembly Dependence. The depth of penetration of the wave generator into the flexspline was found to be the most important among all assembly factors affecting the kinematic error. Experiments were carried out to study the dependence of the kinematic error on load variation (similar to the previous section), while varying the depth of penetration by re-assembly of the harmonic drive. Figure 10 shows the results as the depth of penetration is varied from reference *depth1* (considered as zero) to *depth4* (3.15 mm). The figure shows that at small loads, the amplitude of the kinematic error decreases as the depth is increased, suggesting one possible assembly mechanism to reduce the kinematic error in harmonic drives. On the other hand, large depth of penetration of the wave generator into the flexspline increases the teeth surface contact, inducing tooth wear and possibly other side effects. So, further studies on the effects of large teeth surface contact need to

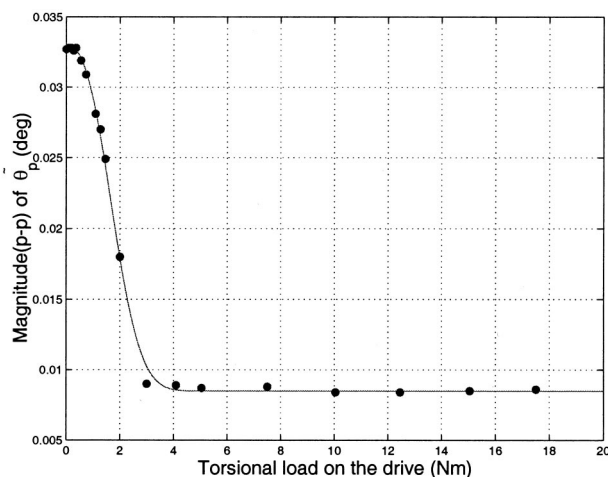


Fig. 9 Dependence of the kinematic error on torsional load

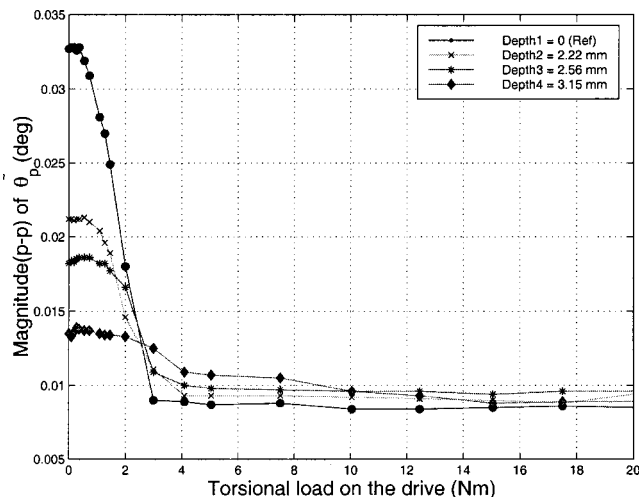


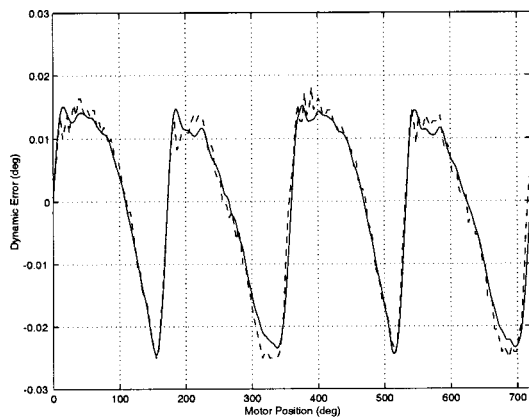
Fig. 10 Dependence of the kinematic error on assembly

be carried out before an appropriate depth of penetration minimizing the kinematic error amplitude could be suggested. Figure 10 also shows that for low torsional loads, the higher the depth of penetration, the smaller the rate of decrease of magnitude of the kinematic error, which could be attributed to an increase in pre-load on the drive due to an increased depth. After a critical load of about 7.5 Nm, the kinematic error amplitude for all depths of penetration exhibits the same constant amplitude, which is in agreement with the conclusions of the previous section.

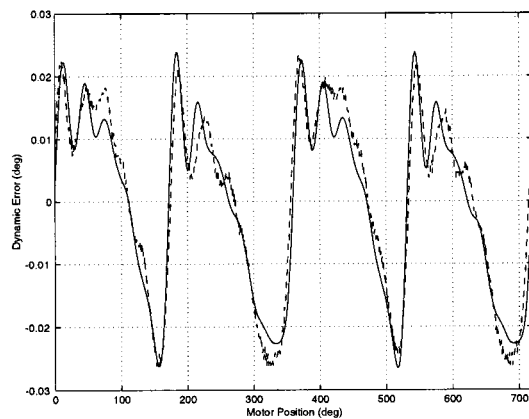
3.3 Angular Speed Dependence. This section presents experimental and simulation results illustrating the dependence of kinematic error on angular speed. The results of simulations of kinematic error carried out at different speeds are shown in comparison with experimental results to illustrate, in particular, the contribution of the flexibility-induced component of the kinematic error. It is noted that the simulation carried out in this section uses the equations of motion (6)–(7) used in the previous simulation in this paper in which the constant stiffness value 7160 Nm/rad is obtained from the experimental stiffness curve of the drive from Fig. 13. In addition, the simulation in this section uses the pure component of the kinematic error with the Fourier representation (8) and coefficients of Table 2.

The simulation and the experimental kinematic error profiles obtained at average speed of 120 rpm with constant input torque applied to the motor are shown in Fig. 11(a). The kinematic error profile clearly has a flexibility-induced component in addition to the pure component. The simulation and experimental results show a very close match, except at the peaks where slight variation is observed. This is most likely due to the unmodeled friction effects at the teeth level.

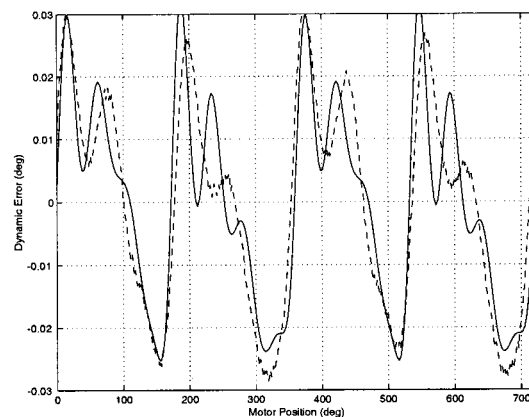
Figure 11(b) shows simulation and experimental profiles of the kinematic error at an average speed of 300 rpm. We first observe that there is an acceptable match between the simulation and the experimental results indicating the suitability of our dynamic model. The small discrepancy between the simulation and experimental profiles is most likely due to unmodeled dynamics, particularly friction. In comparison with the profiles of Fig. 11(a), we notice that as the speed increases, higher frequency components in the kinematic error profile start to manifest. This is explained by the following scenario. At increased speed, the pure component of the kinematic error acts as an exciter inducing vibrations. The frequency of excitation is twice that of the motor speed, i.e. 600 rpm. Hence, the dynamic effects including variation of load position, velocity, acceleration, and corresponding dynamic coupling, are much more dominant than those in the previous lower speed case. Consequently, the flexibility-induced component of the ki-



(a) Speed = 120 rpm



(b) Speed = 310 rpm



(c) Speed = 500 rpm

Fig. 11 Dependence of the kinematic error on speed: Simulation (solid) and experimental (dotted) results

nematic error has a more notable contribution in the overall value of the kinematic error, as confirmed in Fig. 11(b).

Figure 11(c) compares experimental and simulation profiles of the kinematic error when speed is increased to 500 rpm. Again, from the experimental profile, we observe that at higher speeds, the flexibility-induced component has a more pronounced contribution in the overall value of the kinematic error compared to the lower speed cases of Fig. 11(a) and (b). It is noted that even though simulation and experimental profiles share similar trends, we could start noticing more discrepancy between them compared

with the previous two lower speed cases. This is due to unmodeled friction on one hand as observed in the previous two cases, and to the possible change in the profile of the pure component of the kinematic error on the other hand. The latter, as we have shown above, is sensitive to torsional load, which in this case is originating from the higher acceleration inducing higher dynamic loads. It follows that the profile of the pure component of the kinematic error used in the simulation of Fig. 11(c) is slightly different from the actual pure component in terms of peak-to-peak magnitude and profile. This indicates that the dynamic model proposed in (6)–(7) could be made even more accurate by accommodating load sensitivity information of the pure component of the kinematic error (Fig. 9), a task beyond the intended scope of this paper.

4 Conclusions

In this paper we presented new theoretical and experimental results that give a better appreciation of the kinematic error in harmonic drives. A dedicated research Harmonic Drive Test Apparatus was used to carry out the reported experimental results. We first showed that the kinematic error could be decomposed into two components. The first component consists of a basic component representing the “pure” kinematic error. The second component results from inherent torsional flexibility in the harmonic drive gear. The decomposition of the kinematic error into a basic component and a flexibility-related component was demonstrated experimentally. We characterized the dependence of the kinematic error on inertial load, gear assembly, and angular velocity, suggesting new insight in the behavior of the kinematic error under different operating conditions. The proposed mathematical model of the Harmonic Drive Test Apparatus incorporating our model of kinematic error captured reasonably well the dependence of the kinematic error on speed of rotation. But an even more accurate model for a variety of motion conditions could be obtained by incorporating in our model (i) the experimental findings in this paper concerning the dependence of the pure component of the kinematic error on load, and (ii) a model of friction. These two tasks are of a scope that will require a separate study. The results of this paper would be very useful in the mechanical design of harmonic drives as well as a source of accurate dynamical models of harmonic drives for the purpose of devising precision motion control laws.

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Appendix:

A Description of the Harmonic Drive Test Apparatus.

The dedicated apparatus for experimental examination of the kinematic error in harmonic drive gears is reported in Hejny and Ghorbel [19] and is briefly described here. The system is composed of a servo motor, a harmonic drive unit (HDC-40 with reduction ratio 50), and an inertial load. Different sensors are used to measure various system states. The motor position is monitored by a rotary encoder (resolution=0.045°), the load position is measured by a laser rotary encoder (resolution=0.0018°), and the load torque is measured with a DC operated non-contact rotating torque sensor. Figure 12 shows the photograph of the system. As can be seen, the system operates in the vertical plane. Hence, there are no errors in the assembly of the drive because of possible deflection due to gravity. The system is designed with

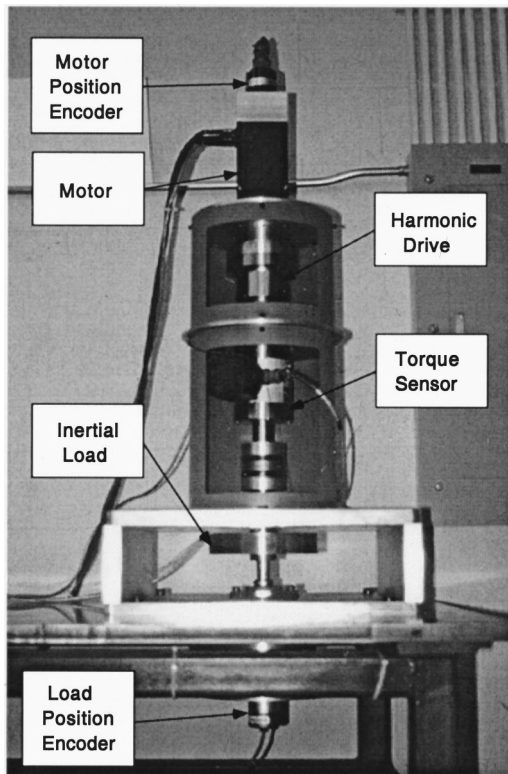


Fig. 12 View of the Harmonic Drive Test Apparatus

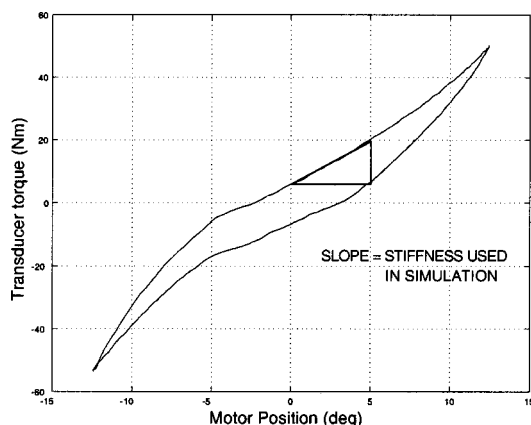


Fig. 13 Steady state stiffness curve of the harmonic drive system

high stiffness so as to avoid assembly deformations when the torque load is applied on the drive. An additional attachment (see Fig. 8) is developed to load the drive with constant torque. Also, an arrangement is provided to lock the output shaft to carry out stiffness measurement experiments. The data acquisition in the experimental system is accomplished through the use of a

dSPACE digital signal processing (DSP) board (dSPACE [21]) hosted in a personal computer with data manipulation and display software.

B Stiffness Curve of the Harmonic Drive. The stiffness curve of the drive is obtained by locking the output shaft (connected to the flexpline). The rotation of the motor and the output of the torque sensor are monitored to obtain the stiffness curve presented in Fig. 13.

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